Conjugate Dirichlet likelihood-learning trajectory

Where the first study fused fixed beliefs in a single round, the second records the KL trajectory produced when a single agent updates the likelihood of its shared world.

Conjugate Dirichlet... (part 2)

The design is a categorical source-mechanism analogue of the language-acquisition mechanism discussed by Friston et al. (Friston et al. 2024).

Conjugate Dirichlet... (part 3)

Each configured seed runs an agent that learns the likelihood of its shared world by conjugate Dirichlet updates over 24 count batches, the count update eq. 18. The recorded trajectory is $\mathrm{KL}(\text{true } A \,\|\, \text{learned } A)$ before each batch —

Conjugate Dirichlet... (part 4)

it starts at the flat-prior value and declines monotonically toward zero as the configured conjugate counts accumulate.

Conjugate Dirichlet... (part 5)

The KL here is the same divergence whose $\alpha \to 1$ Rényi limit is established by Lemma 3 (eq. 6), so the learning curve and the recovery limits measure the same object.

Conjugate Dirichlet... (part 6)

The endpoint summary is: initial KL (flat prior), 3.4231; final KL after 24 batches, 0.0027; and total KL reduction, 3.4204.

Conjugate Dirichlet... (part 7)

The trajectory contains 25 ordered learning steps per seed. Pointwise 95% intervals resample the $n = 480$ independent seeds, and the computed monotone-decreasing indicator is Yes.

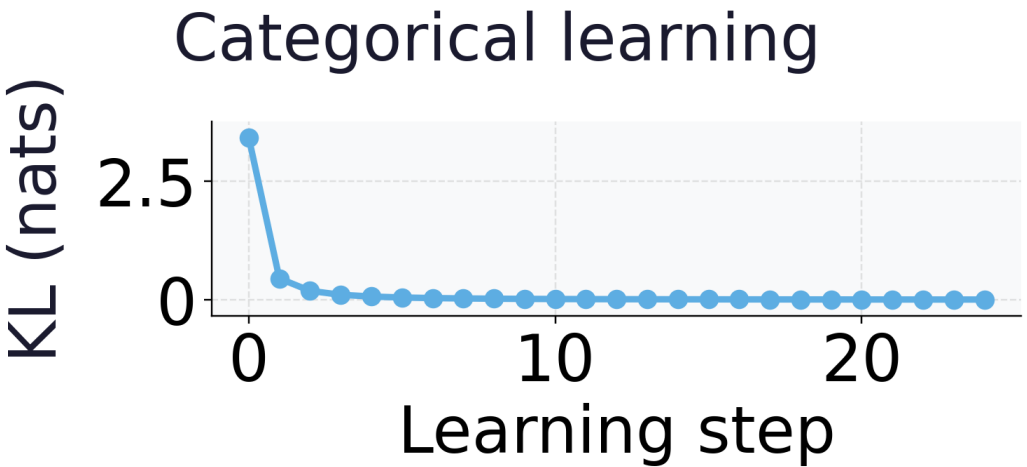
Conjugate Dirichlet... (part 8)

The observed decline to a final KL of 0.0027 is the bounded result: under the tested count schedule, the learned likelihood moves toward the true generative likelihood as conjugate counts accumulate.

Conjugate Dirichlet... (part 9)

This finite trajectory characterizes the configured update and is not a convergence-rate result for arbitrary data-generating processes.

Conjugate Dirichlet... (part 10)



Untagged PDF derivative | HTML reader

Conjugate Dirichlet… (part 11)

KL(true A || learned A), in nats. Circles / solid: seed mean. The ordered x-axis is the Dirichlet count batch.

Conjugate Dirichlet... (part 12)

Pointwise 95% percentile-bootstrap interval over independent configured seeds. Ordered steps are not independent replicates.

Conjugate Dirichlet... (part 13)

Initial mean: 3.423 nats. Final mean: 0.00269 nats. Steps: 25. Seeds: 480.

Conjugate Dirichlet... (part 14)

Source monotonicity check: True.
This is a source-mechanism analogue,
not exact protocol replication.

Conjugate Dirichlet… (part 15)

fig. 14 plots the full learning curve and its CI band.